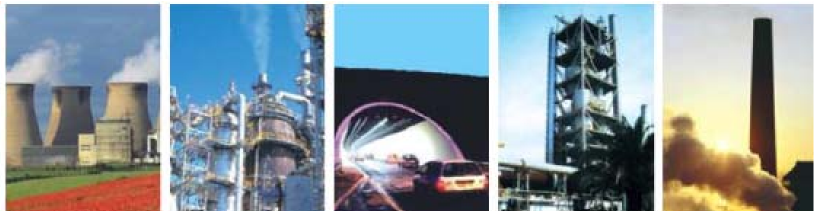

GCEM-40E
Extractive Multi-Gas Analyser

Installation, Operation and Maintenance Manual

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Contents

1 Introduction

CODEL International Ltd. is a UK company based in Bakewell, Derbyshire, specialising in the design and manufacture of high technology instrumentation for monitoring combustion processes and atmospheric pollution emissions. The company philosophy is simply, to design well-engineered, rugged and reliable equipment, capable of continuous operation over long periods with minimal maintenance.

While every effort has been made to ensure that this manual is free from errors, CODEL supply all information without warranty.

Please take a few minutes to read this manual before proceeding with installation. It is designed to provide an overview of the analyser and its capabilities so that the information given later will be more easily understood.

2 GCEM-40E Analyser

The CODEL GCEM-40E analyser has been developed specifically to measure gaseous pollutants in small, inaccessible or high temperature exhaust ducts that are unsuitable for conventional in-situ measurement. It is also useful where it is necessary to measure low-level emissions. A complimentary opacity/dust density measurement can also be included where required.

2.1 Basic principles for gas analysis

The GCEM-40E analyser is a simple extension of the well-established CODEL GCEM-40 non-sampling in-situ analyser.

The basic CODEL GCEM-40 analyser uses an in-situ measurement chamber held directly in the flue gas duct or stack. It uses infrared absorption spectroscopy to measure up to five pollutant species in a single self-contained unit. Flue gases diffuse into the optical measurement path of the analyser where they are analysed.

Such non-sampling in-situ technology enables good, reliable measurement to be obtained even in areas of extreme dust content such as a pre-ESP (ElectroStatic Precipitator), but there are limitations. Measurement with very high or highly variable flue gas temperatures can limit the application of this highly attractive technique.

In such unacceptably high temperature applications, or where the anticipated pollutant levels are too low for accurate analysis using a conventional GCEM-40 in-situ analyser, or where the physical size of the duct limits the use of an in-situ probe, the GCEM-40E provides the obvious alternative. The problems associated with in-situ measurement are eliminated by controlling the absolute stability of the measurement chamber inside a separate but close-coupled free-standing cabinet.

The GCEM-40E incorporates a transceiver unit that transmits a beam of infrared energy through a temperature-controlled measurement cell to a mirror where the beam is reflected back to the transceiver. Analysis of the received energy beam by NDIR technology provides a continuous measurement of the levels of selected emissions within the flue gas. An integral temperature-controlled weather cover enables operation over a 70^{o}C ambient temperature range.

An optional integral Zirconia analyser measures excess oxygen within the sampled gas chamber.

It is worth noting that many conventional extractive systems require the sampled gas to be cleaned and dried to a very high standard prior to analysis invariably resulting in a high-maintenance demand. Such elaborate pre-conditioning is not required by the GCEM-40E analyser. Process gases are continuously drawn through a coarse filtered probe located in the exhaust duct and transported to the measurement chamber via a short heated sample hose - without further conditioning. Figure 1 illustrates the arrangement.

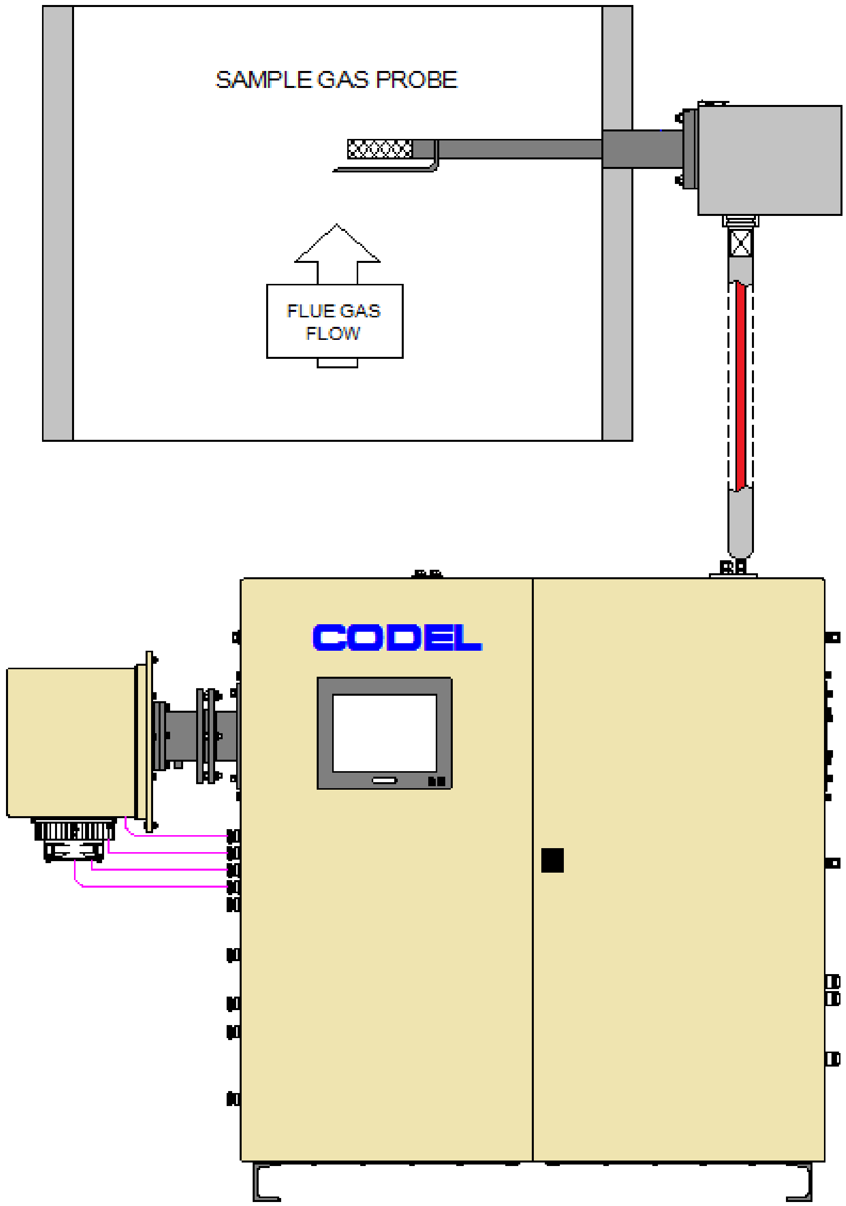


Figure 1 – GCEM-40E Arrangement

2.2 Basic principles for optional opacity/dust density

Smoke and solid emissions have for a long time been recognised as major atmosphere pollutants, particularly since such emissions from stacks are clearly visible to an observer.

There has been a requirement for monitoring and quantifying these emissions for some time and a variety of instruments have been marketed throughout the world for this purpose but instruments in the past have generally proved to be either unreliable – falling rapidly into disuse – or to be so expensive and complex as to only be affordable by the very large users. The CODEL Model DCEM-1000 Opacity Monitor that is optionally integrated into the GCEM-40E seeks to overcome these problems by providing a reliable, simple to use instrument with low maintenance requirements.

It is a single-pass cross-duct optical transmissometer with a transmitter mounted on one side of the duct and a receiver mounted diametrically opposite. It measures the amount of visible light attenuated by particles in its optical path and this reduction in light can be expressed in units of either Opacity or Dust density (after accurate comparative dust testing by others).

High efficiency air purges (air curtains) continuously protect the transmitter and receiver from optical contamination and from overheating by hot process gases.

A typical arrangement is shown in Figure 2.

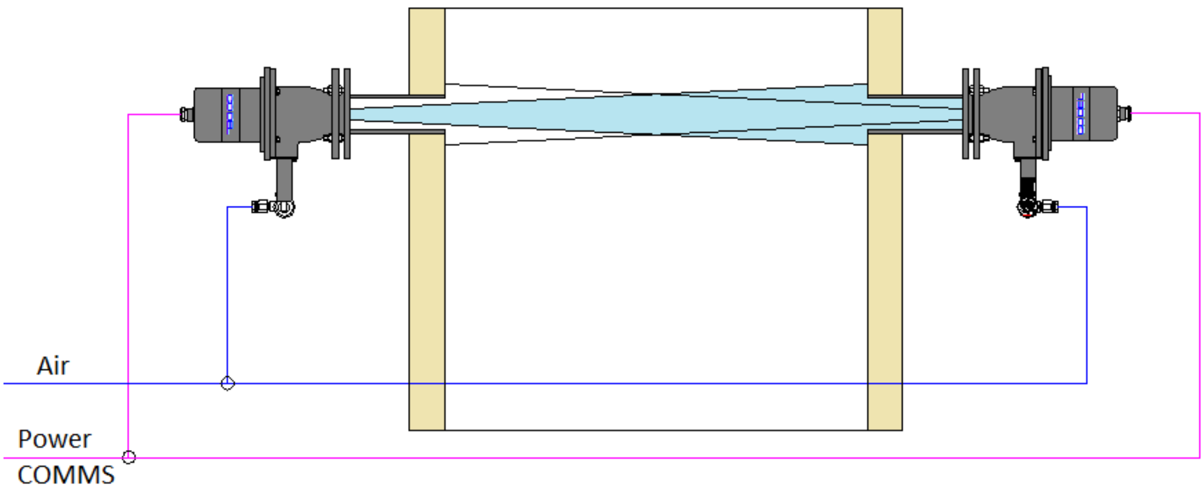


Figure 2 – Optional DCEM-1000 Arrangement

2.3 Measured species

The GCEM-40E analyser can provide measurements of combinations of the following species:

Species	Units	Minimum Range	Maximum Range
CO	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
SO2	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
NO	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
NO2	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
NOx	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
HCl	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
CH4	ppm, mg/m^3 & mg/Nm^3	0-100ppm	0-6000ppm
CO2	%	0-25%	0-25%
H2O	%	0-25%	0-25%
O2	%	0-25%	0-25%
Opacity	%	0-20%	0-100%
Dust density*	mg/Nm^3	0-100mg/Nm^3	0-10000mg/Nm^3
Temperature	°C	0-200°C	
Pressure	kPa	0-160kPa	

* The range and accuracy of dust density measurement is highly dependent upon the measured path length and accurate comparative dust testing. Please request assistance if unsure.

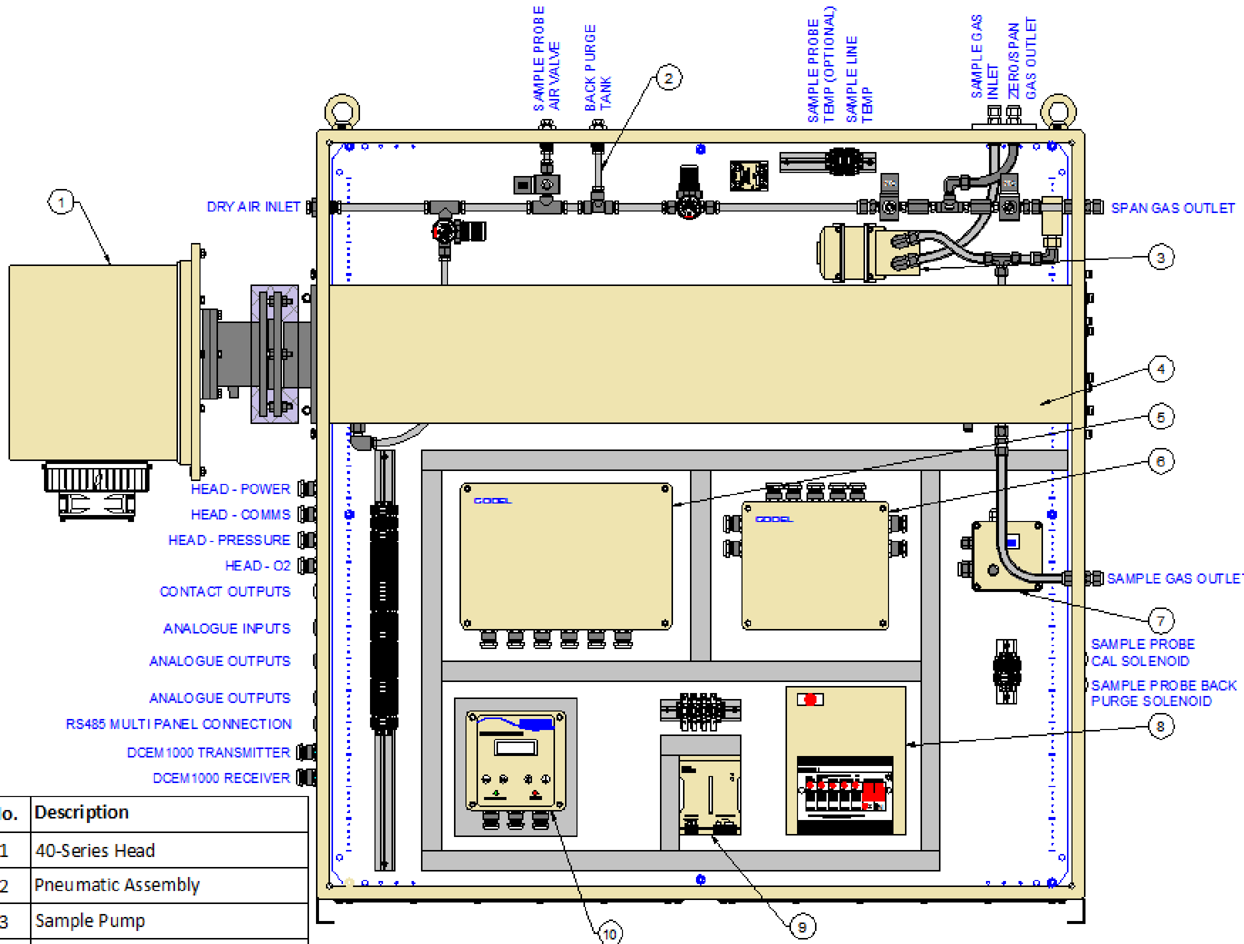
Please note also that the system is limited to measure:

- O2, cell temperature and cell pressure
- plus opacity OR dust density
- plus five gaseous species

2.4 Main system components

2.4.1 GCEM-40E panel

This is the main panel where gas analysis takes place.



No.	Description
1	40-Series Head
2	Pneumatic Assembly
3	Sample Pump
4	Heated Tube Assembly
5	Station Control Unit
6	Heater Control Unit
7	O2 Signal Processor/Display
8	5 Way Consumer Unit
9	24V, 480W Power Supply
10	DCEM1000 Signal Processor/DDU

Figure 3 – GCEM-40E panel internal arrangement

2.4.2 Sample gas probe

The flue gases at the sample inlet are often at a high temperature and/or carry a high dust burden. Prior to analysis they must be cooled and the majority of the dust removed. This is a simple process if the gas is continuously maintained above its dew point.

A sample of the exhaust gas is collected via a conventional sample probe with a heated external filter (See Figure 4). The heated filter can be easily removed for periodic cleaning by removing the cover and then loosening and turning the filter clamp.

BEWARE – if this is attempted with the filter powered-up, it will be at a very high temperature and will cause burns to naked hands.

Sample probes must extend into the duct far enough to collect a sample that is representative of the average composition across the duct.

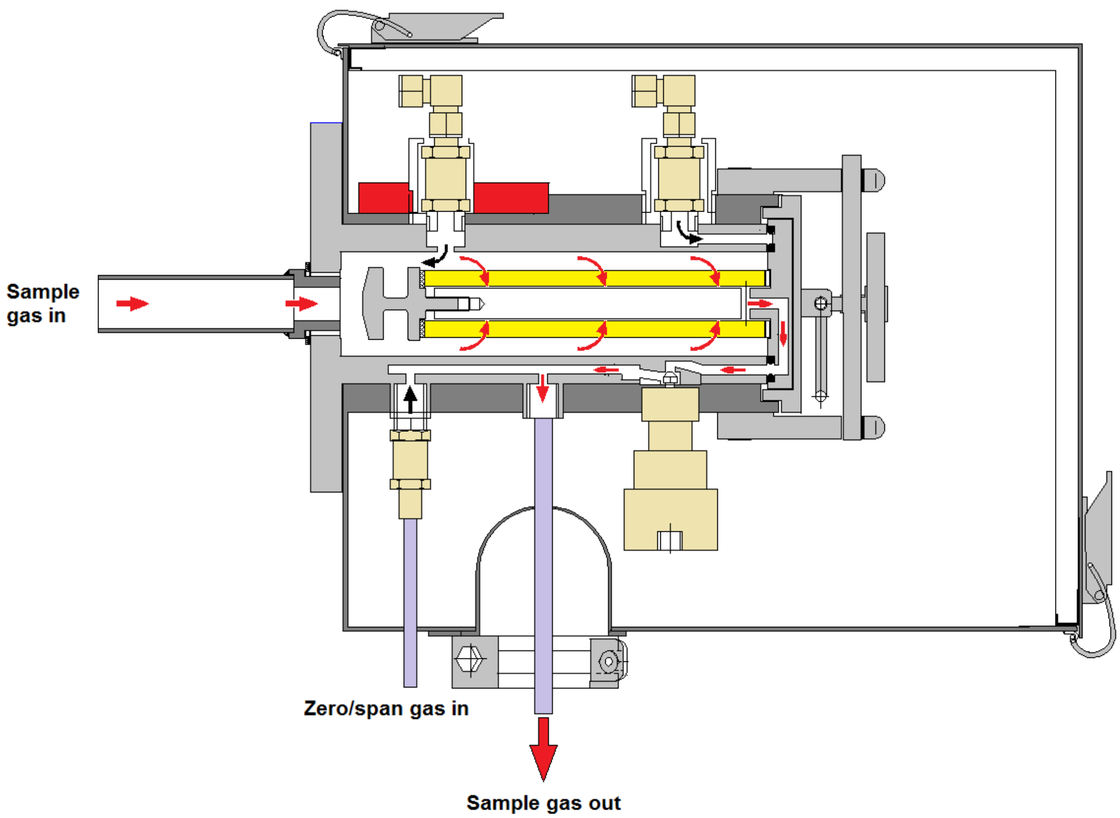


Figure 4 – Sample gas probe

2.4.3 Heated sample line

For optimum performance, the heated sample line must as short as possible. Extended lines demand increased power consumption, create unacceptable delays in the response of the system, and present more opportunity for failure.

Minimum specifications for the sample line are:

No of cores	2 (Sample and calibration gas)
Core size	10mm OD 8mm ID
Core material	PTFE
Operating temperature	180°C
Power per metre run	80W

2.4.4 Heated sample pump

An integral hot gas pump samples flue gases via the sample probe and heated sample line at typically 3 - 5 l/min. After analysis the sampled gas is vented from the side of the cabinet. It should ideally be piped to the nearest drain. NOTE: No restriction of flow must occur.

2.4.5 Heated measurement chamber

The insulated measurement chamber is controlled to a uniform temperature of 180°C. Control over 5 zones is fully automated and requires no operator intervention.

2.4.6 Multi-gas analyser

The GCEM-40E transceiver unit is mounted outside the main cabinet and fits directly on the end of the heated measurement chamber. An integral temperature-controlled weather cover enables operation over a 70°C ambient temperature range (typically -20 to +50°C).

2.4.7 Oxygen analyser

A compact Zirconia analyser is fitted into the measurement chamber to measure excess oxygen in the sampled gas.

2.4.8 Dust/opacity transmitter and receiver (Optional)

The optional DCEM-1000 transmitter uses a high intensity light emitting diode (LED) as the light source. An internal detector monitors the transmitted light intensity. A lens in the front of the enclosure focuses the light from the LED across the duct.

Current to the LED is modulated at a frequency of 512Hz. The wavelength of the light emitted is 637 nanometres which is in the red part of the visible spectrum and the source intensity is sufficient for duct widths up to six metres to be accommodated.

The receiver consists of a silicon cell detector and its associated electronics housed in an identical enclosure to the transmitter. A lens focuses the light received onto the detector.

Because the light from the transmitter is modulated, the receiver is able to measure ambient light levels, and the levels compensated for within the signal processor.

Both units are housed in fully-sealed (to IP65) epoxy-coated aluminium enclosures.

2.4.9 Dust/opacity signal processor and Data Display Unit (Optional)

The integrated dust/opacity signal processor and DDU (Data Display Unit) control the optional DCEM-1000 and permit local access and interrogation.

2.4.10 Station Control Unit (SCU)

At the heart of this analyser is the Station Control Unit (SCU) that provides power and communication to all equipment within the system for normalisation routines and analogue/contact/serial inputs and outputs. It manages the sequencing of integral solenoid valves for the automatic injection of zero and span calibration gases into the analyser. It also controls the sample line and measurement zone temperatures.

2.4.11 Human Machine Interface (HMI) (Optional)

On the front of the cabinet can be fitted an HMI with 12” touch-screen display. The principal function of this HMI is to provide measured and historical data in multiple formats for operators. It also allows access to configuration and diagnostic data for commissioning and trouble-shooting.

2.4.12 Compressed air filter/drier unit

For automatic calibration, the GCEM-40E-SCR uses compressed air as a convenient zero gas – however, plant compressed air ***must*** be clean, oil-free and dry to -20°C. Where this is not available, a combined compressed air filter/drier unit can be supplied by CODEL.

Clean dry compressed air must also be supplied for continuous purging of the optional opacity/dust density monitor.

2.4.13 SmartCom Software

This is a CODEL software suite ideally suited for commissioning. It enables all analyser output and diagnostic data to be accessed, logged and configured via a separate laptop PC.

2.4.14 SmartCEM Software

This is the operational CODEL CEMS software suite that continuously presents measured data in the format demanded by legislation. It provides customer-friendly access to both live and long-term historic data via the integral touch-screen PC.

3 Measurement Principles - Gas

3.1 Measurement chamber

Gas analysis is made within a temperature-controlled chamber inside the free-standing cabinet. This chamber consists of two major parts – one is the chamber through which the hot gases flow and analysis takes place. The second is a dead zone separating the delicate components of the transceiver from the hot gas in the measurement chamber. The dead zone is permanently purged with zero gas to flush away any pollutants that might build up in this chamber.

Measurements are made using NDIR (Non Dispersive InfraRed) technology. A beam of infrared energy is transmitted down the probe from the sensing head to a mirror at the end of the measurement chamber from where it is reflected back to the sensing head where it is spectro-analysed to determine the concentration of specific pollutants in the flue gas.

3.2 Measurement Elements

An infrared beam is generated by a small thermal source within the transceiver. Radiation from this source is focused by a lens onto the mirror mounted at the far end of measurement chamber. The reflected beam is focused by a second lens in the transceiver onto a highly sensitive infrared detector. Immediately in front of the detector is a gas wheel that holds a combination of optical filters and gas cells designed to isolate specific infrared wavebands required for the measurement of the defined gas species. This wheel is rotated by a stepper motor at a constant speed of 1Hz. As each filter sweeps through the infrared beam, the on-board processor digitises the detector output to produce a detector value ‘D’.

The number and types of optical filter used depends upon the gas species being measured. However each gas species is characterised by two detector measurements one of which is a ***live*** or measurement channel uniquely sensitive to the gas being measured (Dmeasure); the other is a ***reference*** measurement at a waveband insensitive to the measured gas (Dreference). From these two values, specific to each gas species, the gas concentrations can be calculated.

3.3 Detector Operation

The detector is a lead selenide photoconductive element. To achieve the sensitivity required for accurate gas analysis the element must be cooled to a temperature of $-25^{o}C$. This is achieved by an encapsulated thermoelectric Peltier cooler with an integral thermistor to monitor the element temperature.

The on-board processor continuously monitors the thermistor temperature and regulates the power to the Peltier cooler to achieve the required stable temperature.

3.4 Stepper Motor Control

The supervisory processor develops a frequency signal which is used to drive the stepper motor and gas wheel. Accurate timing of this signal ensures that the wheel rotates at precisely 1Hz. By counting the pulses in the frequency drive to the motor the processor knows exactly when each optical filter is passing through the infrared beam and can conduct the detector digitisation process to obtain the detector signals necessary for the calculation of the gas species concentrations.

3.5 Calculation of Gas Concentration

Every second, with each rotation of the gas wheel, a series of detector measurements are made resulting in values of Dmeasure and Dreference for each gas species being measured. A parameter which we define as Y is calculated for each gas according to the relationship:

$$Y = 80000 - SC \times Dmeasure/Dreference$$

Where:

SC is a calibration constant designed to make Y = 0 when the gas concentration is zero.

This ‘Y’ value is a unique, but non-linear function of gas concentration.

Having calculated the ‘Y’ value for each species, the processor then applies a linearising algorithm to the ‘Y’ value to produce a value of gas concentration in ppm (parts per million by volume).

3.6 Effects of Gas Temperature

Gas density and the infrared absorption spectra of the gases are both influenced by flue gas temperature. For an accurate determination of gas concentration it is essential to know precisely the temperature of the gas. In the GCEM-40E-SCR the measurement chamber is controlled to a high accuracy to eliminate any affects due to temperature variation.

3.7 Output Data Normalisation

For emissions monitoring purposes it is vital that operators are not allowed to dilute the measured value by introducing excess air, or by heating the gas prior to measurement.

It is common practice therefore to present gas concentration data not only in ppm, but also in mg/Nm^3 (milligrams per Normal metre cubed). For conversion from ppm to mg/Nm^3 it is necessary to have knowledge of the gas temperature, pressure, moisture content dilution air content and molecular weight. Each variable parameter must be measured to allow the gas concentration to be corrected to standard gas conditions – normally:

0^oC temperature,
1 atmosphere pressure,
zero moisture content,
a pre-defined level of excess O2 or CO2 (representing a typical level of dilution air for that process).

This system is fitted with an integral absolute pressure sensor which is continuously monitored by the on-board processor. The sensing head is also equipped with an additional measurement channel for H2O thus enabling complete data normalisation to be provided without the necessity of acquiring additional data from other sensors and analysers.

3.8 Analyser Temperature Control

Stable operating temperatures are essential for accurate measurements using infrared technology. The GCEM40 Series sensing head is equipped with state-of-the-art temperature control technology built around the incorporation of an integral fully IP66 sealed weather cover with thermoelectric heating/cooling capabilities.

3.9 Auto Weather Cover Control

The auto weather cover AWC enables the sensing head control temperature to be maintained within +/-0.1^{o}C over an ambient temperature range of -20 to +50^{o}C.

3.10 Gas Wheel Temperature Control

The gas wheel holds the optical filters and gas cells necessary for the measurements. These items are temperature dependent and must be held at a constant temperature for accurate gas analysis to be made. A heater plate positioned close to the wheel is controlled by the on board processor to ensure a stable operating temperature for the gas wheel. This temperature is set 2^{o}C above the sensing head control temperature maintained by the AWC.

3.11 Thermistor Control

In to achieve the detector sensitivity required for these measurements, the detector element temperature must be maintained at a level of -25^{o}C. This is achieved by incorporating a thermoelectric cooling element and thermistor temperature sensor in the detector package. Power to the cooler is regulated by the processor to maintain a constant detector element temperature, as measured by the integrated thermistor.

3.12 Analyser Calibration

The system is designed with an inlet port that allows calibration gas to be injected into the measurement chamber thereby flushing out the sampled flue gases and replacing them with calibration gas in order to maintain and verify calibration accuracy. Two solenoid valves operated by the sensing head processor are used to control the flow of zero gas (dry compressed air) and span gas (protocol cylinder gas).

A zero calibration is performed automatically every 24 hours and on demand manually via the HMI. Regular auto zero calibration ensures control and correction of any long term calibration drift.

Span verification checks can be activated on demand from via the HMI after ensuring that the correct protocol gas cylinder is connected to the span input solenoid on the probe.

4 Measurement Principles – Opacity/Dust Density

4.1 Optical Transmissivity Measurement & Solid Content

Since the aim of all optical transmissivity measurements is to assess the level of solids emission, it is important to consider the relationship between various optical measurements and the particulate content of the gas. It must first be recognised that the optical methods can only be applied if the particulate size is small enough for the mass of the solids to be considered as spread evenly over the cross section of the flue. In general this limits the optical techniques to particles of 20 micron or less diameter.

Providing the particle size criterion is met, there are certain very specific relationships between the optical measurements and the particulate concentration.

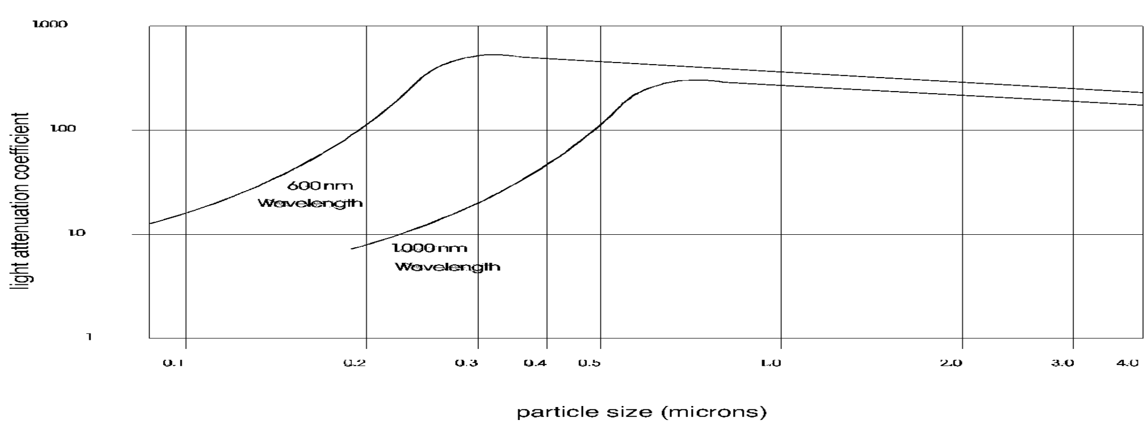


Figure 5 – Relationship between particle size and attenuation

4.2 Extinction Coefficient and Beer Lambert Relationship

The attenuation of light of a narrow waveband - such as visible or photopic light by fine particulate matter is very specific and is expressed mathematically by the Beer Lambert Relationship:

Transmittance T = exp (-k.n.a.l)

Where: k is the extinction coefficient of the particles
n is the number of particles per unit volume
a is mean projected area of the particles
l is the path length through the gas

Thus Opacity = 100 [1 - exp (-k.n.a.l)] %

The extinction coefficient k determines precisely how much light is attenuated by the particles and depends totally on the physical and chemical nature of the particles. For a given flue gas, the particulate type and size distribution should be relatively constant so that the opacity measurement may, for a given path length l, be related to the particulate concentration n.

4.3 Optical Density & Extinction

Optical density and Extinction are different names for the same parameter. This parameter is useful in that assuming the nature and size distribution of the particles are constant, for a given path length, it is directly proportional to the particulate concentration n, and hence to particulate mass emission.

Optical density is expressed:

Optical Density = $\log_{10} 1/T$

and, since T = exp (-k.n.a.l),
Optical density = k.n.a.l $\log_e 10$

The following table provides a simple conversion from measurements in terms of opacity or transmittance to optical density/extinction.

Opacity %	0	10	20	30	50	70	90
Transmittance	1	0.9	0.8	0.7	0.5	0.3	0.1
Opacity ($\mathbf{Log_{10}}$)	0	0.05	0.1	0.15	0.3	0.52	1.00

4.4 Mass Emission

While it is possible to provide an output for opacity monitors in terms of optical density/extinction, conversion of this parameter into a measurement of mass flow emission, in mg/m^3 for example, requires an empirical calibration to be made for that particular installation and flue condition. This must be achieved by calibrating against specific measurements of solids emissions, which would have to be made by independent iso-kinetic sampling. The resulting calibration would hold only for that particular flue and would need to be rechecked at least every six months by further sampling comparisons.

4.5 Opacity/Ringelmann Correlation

Ringelmann is a measurement of opacity at stack outlet. If the opacity measurement were also made at stack outlet the measurements would be exactly equivalent. However, it is normally more practical to measure opacity at some other point for ease of access. In this situation, the measurements will differ since the duct diameter at the measurement point will differ from that at the stack exit, and the measurement depends upon the length of the optical path.

Opacity at the point of measurement

$$O_m = 100\,[1 - \exp(-k.n.a.l_m)]\ \%$$

where m is the stack diameter at the point of measurement.

Opacity at stack exit

$$O_x = 100\,[1 - \exp(-k.n.a.l_x)]\ \%$$

where lx is the stack exit diameter.

4.6 Calibration for Dust Measurement

For Opacity Monitors to provide a measure of the dust density, the relationship between opacity and the dust density needs to be established. For a particular measurement point in a particular flue, duct, or chimney, this can only be accomplished accurately by iso-kinetic sampling over a defined period. Over the same period, the opacity monitor will calculate the corresponding average extinction, which is directly proportional to the dust density

The relationship between extinction and dust density may be described as the ‘Dust Factor’.

Dust Density = Fine Dust Density/Extinction

Where:
Fine dust density = mg/m^3 of dust by iso-kinetic sampling
Extinction = average extinction for the same period.

This Dust Factor is input and held within the instrument to convert extinction to fine dust density. Some important points to note when calibrating from iso-kinetic sampling results are given below.

Take care to ensure that the normalisation parameters temperature, oxygen, pressure & water vapour) are set to sensible values, and establish to what standard conditions the sampling results will be normalised to. Iso-kinetic results are automatically brought to ambient temperature as the flue sample cools through the sample line, and in general these are then normalised to ^{0}C (273K). Normalisation to the other parameters is variable, and depends largely upon whether they are being measured at the same time as sampling.

It is reasonable to start with an estimate of the dust factor (see next Section) and ensure that at least the correct temperature is held within the instrument, entered via the keypad. If it is uncertain whether the sampled data will be normalised to the other conditions, they may be taken out of operation by bringing the measured values to the standard levels, e.g., O_2 standard level 3%, value fixed at 3 %, Pressure standard level 101kPa, value fixed at 101kPa, Water Vapour standard level DRY, value fixed at 0 %.

While the sampling is being conducted, record the minutes average and normalised dust value mg/Nm^3 preferably from a chart recorder, if used, (making sure of course, that the current output has been correctly configured). If not available, however, the displayed normalised dust value should be recorded regularly for comparison.

After the sampling has been conducted, the two results may be compared, and the dust factor adjusted if necessary, to bring the calibration to that measured by iso-kinetic sampling.

Take care to compare only data that has been normalised to exactly the same standards.

4.7 Estimate of a Dust Factor

For the model DCEM1000 to calculate the dust density from the opacity data, a Dust Factor needs to be determined by iso-kinetic sampling. However, if there is a delay before sampling can be conducted, an estimate of the dust factor may be used.

Estimated Dust Factor = 2500 / path length in metres

It must be emphasised that this approximation of the dust factor cannot be accurate and that it might be as much as a factor of 2 in error – particularly if the particle size is large.

5 Basic Specification

5.1 Measurements	Operating Principle	NDIR gas filter correlation
	Span	0 to 6000ppm (CO, NO, SO2) 0 to 25% (CO2, H2O)
	Response Time	<200secs
	Accuracy	+/-2ppm or +/-2% of span
	Resolution	1ppm
	Zero Calibration	automatic every 24 hours
	Span Verification	manually on demand
5.2 Compliances	EMC	89/336/EEC compliant
	Low Voltage Directive	73/23/EEC compliant
5.3 Outputs	Analogue	8x 4 to 20mA isolated, 500Ω load. Fully configurable from HMI
	Contact	8x volt-free SPCO contacts. 50V, 1A max. Configurable as alarms
	Data Valid Contact	1 x volt-free SPCO contact, 50V, 1A max, for data valid only
	Serial Data	RS485 Standard MODBUS or RS485 CODEL SmartCEM
	Serial Diagnostics	RS232 CODEL SmartCom
5.4 Inputs	Analogue	1x 4 to 20mA self-powered representing flue gas flow 2x 4 to 20mA configurable from HMI
5.5 Power	System voltage	110/240VAC, 50-60Hz, 1 phase
	Analyser load	1000VA
	Sample probe load	800VA
	Heated line load	Additional as required
5.6 Air	Compressed air	4bar Dry to -20°C Clean (better than 10μm)
	- continuous	20 l/min (35 l/min with DCEM)
	- during calibration	30 l/min (45 l/min with DCEM)
5.7 General	HMI Display	12” monitor
	Keypad	Touch screen
	Construction:	
	- sample probe	316L stainless steel
	- GCEM head	Epoxy coated aluminium
	- cabinet	IP55 Mild steel with RAL 7035 structure powder coating
	Cabinet weight	190 kg
	Ambient temperature	-20 to +50°C
	Max flue gas temp	600°C

6 Preparing for Installation

6.1 Unpacking

CODEL GCEM-40E-SCR analysers are normally protected for transportation using an expanded foam packing material. When unpacking please ensure that smaller items are not discarded with the packing material.

If any items are missing please inform CODEL or your local CODEL agent immediately.

6.2 Basic Analyser Supply

Please refer to contract-specific documents for accurate details.

6.2.1 Standard supplied items

- GCEM-40E-SCR analyser
- Sample probe (material and length to suit the application)
- Heated sample filter
- Cable glands
- Probe site mounting flange – DN65 PN6 (mild steel)
- Gaskets for sample probe (x2)
- Compressed air filter/drier

6.2.2 Optional supplied items

- DCEM-1000 transmitter
- DCEM-1000 receiver
- 2x Probe site mounting flange – CODEL part 900.143A (mild steel)
- Gaskets for DCEM transmitter and receiver (x2)
- Compressed air filter/drier

6.2.3 Customer supplied items

- Heated sample line
- The following cable and piping:

Function	From		To		Minimum specification
GCEM-40E system power	Client's power	Isolator	GCEM-40E panel	Consumer unit	3-Core Cable, 240V, 50Hz, 1ph, 100A
Contact outputs	GCEM-40E panel	Terms 7 - 22	Clients Computer/DCS		8 x pairs 50 VDC 1A max
Analogue inputs	Client		GCEM-40E panel	Terms 23 - 28	3 x Twin Twisted Pairs 500 Ohm MAX
Analogue outputs 1-4	GCEM-40E panel	Terms 29 - 36	Clients Computer/DCS		4 x Twin Twisted Pairs 500 Ohm MAX
Analogue outputs 5-8	GCEM-40E panel	Terms 37 - 44	Clients Computer/DCS		4 x Twin Twisted Pairs 500 Ohm MAX
RS485 to/from next panel	This panel	Terms 45 - 46	Next panel	Terms 45 - 46	RS485 twisted pair axial screened cable. 120Ω characteristic impedence. Max
Sample line power	GCEM-40E panel	Terms 47 - 49	Heated sample line	Fixed tails	Fixed tails
Sample line temperature	Heated sample line	Fixed tails	GCEM-40E panel	Terms 52 - 53	Fixed tails
Probe back-purge valve	GCEM-40E panel	Terms 54 - 55	Probe back-purge valve	Terminals	2-core cable 0.5mm2 min
Probe isolation valve	GCEM-40E panel	Terms 56 - 57	Probe isolation valve	Terminals	2-core cable 0.5mm2 min
Sample probe power	GCEM-40E panel	Consumer unit	Sample probe	Terminals	2-core cable 0.5mm2 min
Plant air supply	Clients Clean Air Supply	Isolating valve	Air dryer	10mm Brass Comp	10mm OD Nylon/PTFE
Autodrains	Air dryer	10mm Nylon Pushfit	Clients drain		10mm OD Nylon/PTFE
GCEM-40E dry air supply	Air dryer	10mm Brass Comp	GCEM-40E panel	10mm Brass Comp	10mm OD Nylon/PTFE
Sample isolating valve	GCEM-40E panel	10mm Brass Comp	Air valve	10mm Brass Comp	10mm OD Nylon/PTFE
Back-purge tank supply	GCEM-40E panel	10mm Brass Comp	Back-purge tank	10mm Brass Comp	10mm OD Nylon/PTFE
Probe back-purge supply	Back-purge tank	10mm Brass Comp		10mm Brass Comp	10mm OD Nylon/PTFE
GCEM-40E sample gas in	Heated sample line	10mm SS tail	GCEM-40E panel	10mm SS Comp	Fixed tails
Zero/span cal gas to probe	GCEM-40E panel	10mm SS Comp	Sample probe valve	10mm SS Comp	10mm OD PTFE
Span gas inlet	Gas bottle	Isolating valve	GCEM-40E panel	10mm SS Comp	10mm OD PTFE
Sample gas vent	GCEM-40E panel	10mm SS Comp	Clients vent		10mm OD PTFE

Where DCEM-1000 analysers are supplied, it will be necessary to supply 2x 3” nominal bore (DN80) sample probe mounting tubes (mild steel).

When calculating cable requirements remember to take into account cable routing relative to equipment location and also the need for conduit, cable trays, supports etc.

Other materials required include cable trays, conduits, ties and associated fixing materials. Consideration should also be given to the possible need for temporary or permanent access platforms, lifting and welding equipment.

7 Installation

PLEASE BE SURE TO READ Section 0

Preparing for Installation ***BEFORE PROCEEDING***

7.1 Selection of Measurement Position

This monitor measures the concentration of gas in a specific section of the duct. Position of the sample probe is therefore critical as consideration must be given to ensuring that results are representative of the bulk gas analysis.

The selected location should comply with the requirements defined in:

ISO 10396:2007 Stationary source emissions – Sampling for the automated determination of gas emission concentrations for permanently-installed monitoring systems.

This provides criteria for assessing significant flue gas stratification – however, as a guide, ISO 10396:2007 comments:

"Usually, the cross-sectional concentration of gaseous pollutants is uniform, because of the diffusion and turbulent mixing. In this case, it is only necessary to sample at one point within the stack or duct to determine the average concentration. A sample gas should be extracted near the centre of the sampling site, positioned one-third to halfway in the stack or duct…"

The next factor to consider is that of flue gas pressure. These analysers are capable of measuring in both positive and negative pressure regions relative to atmospheric pressure. However, for serviceability reasons, it is always safer to choose a negative pressure region if possible.

Positive pressure in the duct will cause hot flue gas to be vented from any open port. Always use protective clothing, especially eye protection, when installing or removing the probe from the duct.

Try to ensure that duct vibration at the selected mounting point is minimal. The gas sample probe is designed to withstand relatively severe vibration conditions, but vibration will inevitably reduce the operating lifetime of the product.

It is vital that the gas sample probe should not interfere with the optical beam of the opacity/dust density measurement. A separation of 300mm along the flow of the duct should be maintained.

Other factors to be considered in selection of the measurement position are those of accessibility and cost of installation. Access from existing platforms or walkways and proximity of existing services are obvious advantages. It is important to ensure continuing ease of access after the installation and commissioning work to facilitate future maintenance. Ensure that there is sufficient space between the mounting location and any nearby obstruction to allow for insertion of the probe.

7.2 Selection of Analyser Cabinet Position

This analyser is too heavy to be man-handled into position. It is supplied with lugs for mechanically lifting into its final position.

Ideally the analyser should be mounted as close as possible to the sample probe to ensure the best response. However, as the maximum weather protection rating for the system is IP55 it would be advisable to mount indoors or at least under cover.

In extreme weather conditions, it may be necessary to provide a heated or cooled shelter.

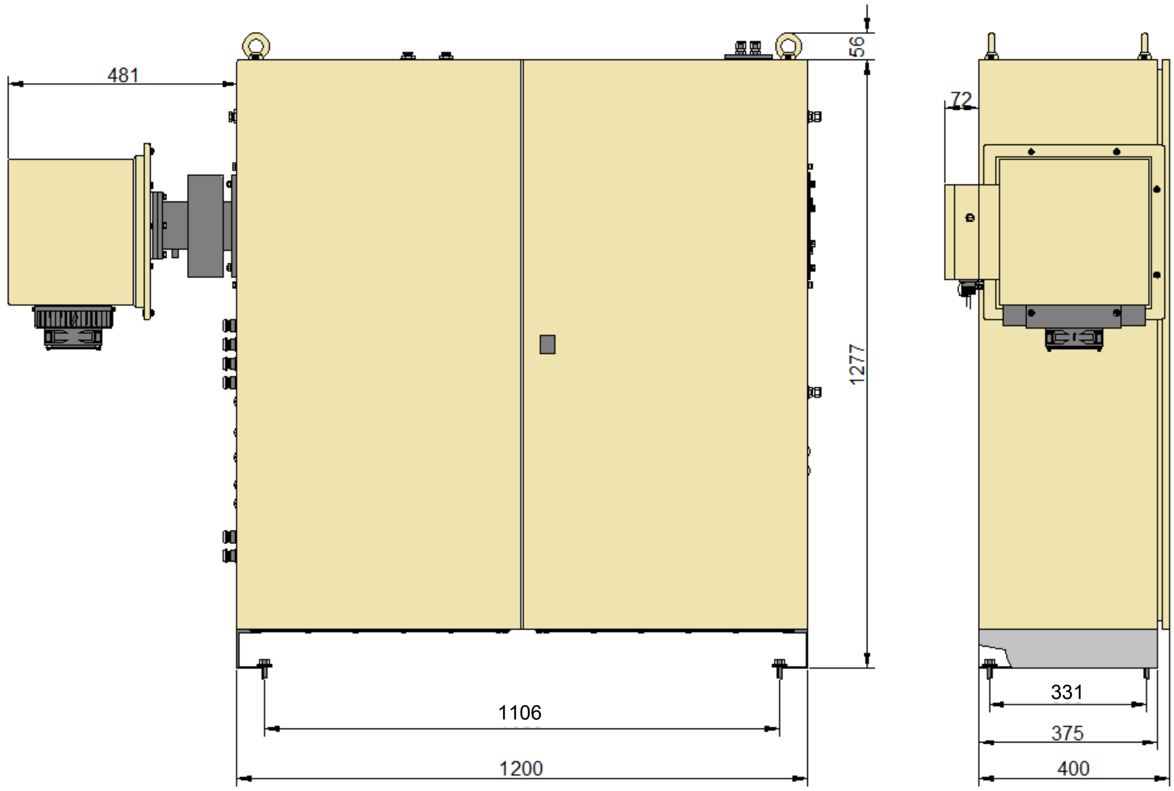


Figure 6 – Cabinet dimensions

Where it is considered necessary to fix the cabinet, it is advised that four M10 fixing holes or studs be fitted (before the cabinet is positioned) in the positions shown. Lift the cabinet into position and then clamp the base using M10 A2 spreader washers.

As this is a heated system, it is vital to provide adequate ventilation by ensuring that all sides of the cabinet (including the back face) are always positioned at least 100mm from any wall.

Although the cabinet is fitted with gland plates, we presume that incoming power cable glands will be fitted by others in the side walls.

7.3 Fitting of Stub Pipes and Mounting Flanges

Before proceeding further, note that under no circumstances should holes be cut into the stack or duct with the plant operational. Positive duct pressures or even brief positive pressure pulses will cause hot flue gas to be vented from any opening. Always wear appropriate protective clothing, including eye protection, when working on open stack or duct ports.

Construct a mounting assembly by welding the DN65 PN6 site mounting flange for the gas sample probe to a stub pipe of 2½" nominal bore (DN65.).

Construct a mounting assembly by welding the 2x CODEL 900.143A site mounting flange for the optional opacity heads to a stub pipe of 3" nominal bore (DN80).

These pipes should be long enough to slip through the duct wall (plus any internal refractory lining), extend through the external cladding and stand off the external cladding by a maximum of 150mm.

Cut a 'slip hole' into the duct or stack at the chosen location. Note that the stub pipe for the sample probe should be mounted between horizontal and up to 10° downwards. This ensures that condensates cannot accumulate inside the sample probe under abnormal conditions.

The stub pipes for the opacity heads must be diametrically in-line.

Please refer to contract-specific drawings for details.

It is vital that the gas sample probe should not interfere with the optical beam of the opacity/dust density measurement. A separation of 300mm along the flow of the duct should be maintained.

For high temperature applications, where there is still excessive radiated heat from the cladding, it might be advisable to fit a heat shield. This should be at least 750x750mm and trapped between the stub pipe and analyser flanges.

7.3.1 Concrete, Brick or Double-Skin Stacks

These instructions are primarily for metal ducts but a great variety of stack dimensions and materials may be encountered. Where this is the case, the stub pipe position and flange orientation should be the same as detailed in Section 7.1 but please consult CODEL about specific methods before commencing installation.

7.3.2 Orientation of Sample Probe

It is necessary to ensure that the in-duct filter guard faces into the direction of gas flow. See Figure 7.

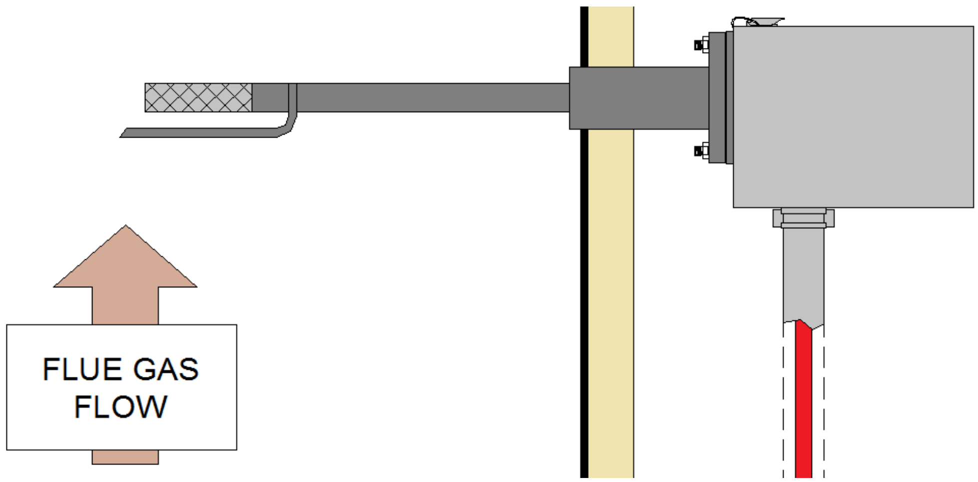


Figure 7 – Orientation of sample gas probe

7.3.3 Installation of Purge Air Supply

The analyser requires a supply of clean dry compressed air as specified. The supplied air drier should ideally be fitted into the air line at a point close to the cabinet – except where the analyser is operating at temperatures below 0°C. Where this is the case, it may be beneficial to mount the air drier unit inside the nearest building. This will minimise the risk of condensates freezing in an undried/unheated air supply.

The compressed air should be supplied at a minimum pressure of 5bar (75psi) to the compressed air inlet point. It must also be Oil free and dry to a dewpoint of -20 degrees or better. Flow restrictors in the system will ensure correct air flow to all the other elements. No adjustment is necessary.

It is important to establish the compressed air connection as soon as possible after insertion of the probe into the duct and before any electrical power is connected to the analyser. This will ensure protection for all temperature-sensitive components.

Note that the supplied air dryer *MUST BE FITTED*. If not fitted, all warranties will be *VOID*.

7.3.4 Installing the Sampling Line

The heated sample line has a pneumatic fitting on the end for attaching to the sample probe. It has a pneumatic fitting and electrical tails for attaching to the top of the cabinet.

Attach the sampling line to the sampling probe at the duct and feed it back to the cabinet. It should be frequently clipped to adequate support structures.

Note that the bend radius of the sampling line should never be less than 0.5m and should fall gradually at all times towards the analyser cabinet. Under no circumstances should the sample line have a low point below the inlet of the cabinet.

The pneumatic fitting should be connected to the sample inlet of the GCEM-40E-SCR analyser and any exposed pipework of the sampling line must be lagged with weatherproof insulation and tape. The tails of the sample line should be connected through the cable gland provided and connected as detailed in the contract-specific connection schedule.

7.4 Installation of Calibration Gases

If required, span verification may be carried out using certified bottled test gases.

Automatic daily zero calibration does not require certified test gases. On that basis, bottled gases may be connected permanently or simply connected as required.

Where these bottles are large enough to cause injury to personnel, safe storage must be provided.

7.5 Wiring Connections

Wiring should only be undertaken by a qualified technician.

Ensure that the power supply is isolated before attempting any wiring.

DO NOT SWITCH POWER ON until all installation is complete, the wiring connections have been checked as correct and the system is ready for commissioning.

7.5.1 Installation and Connection of Cables

Decide routing for all non-power cables. Use common routing wherever possible and install leaving sufficient free-end length to make final connections.

7.5.2 System Connection Schedule – Main System

Typical inter-connections for the system cables are shown in Figure 8.

Please refer to contract-specific drawings for accurate details.

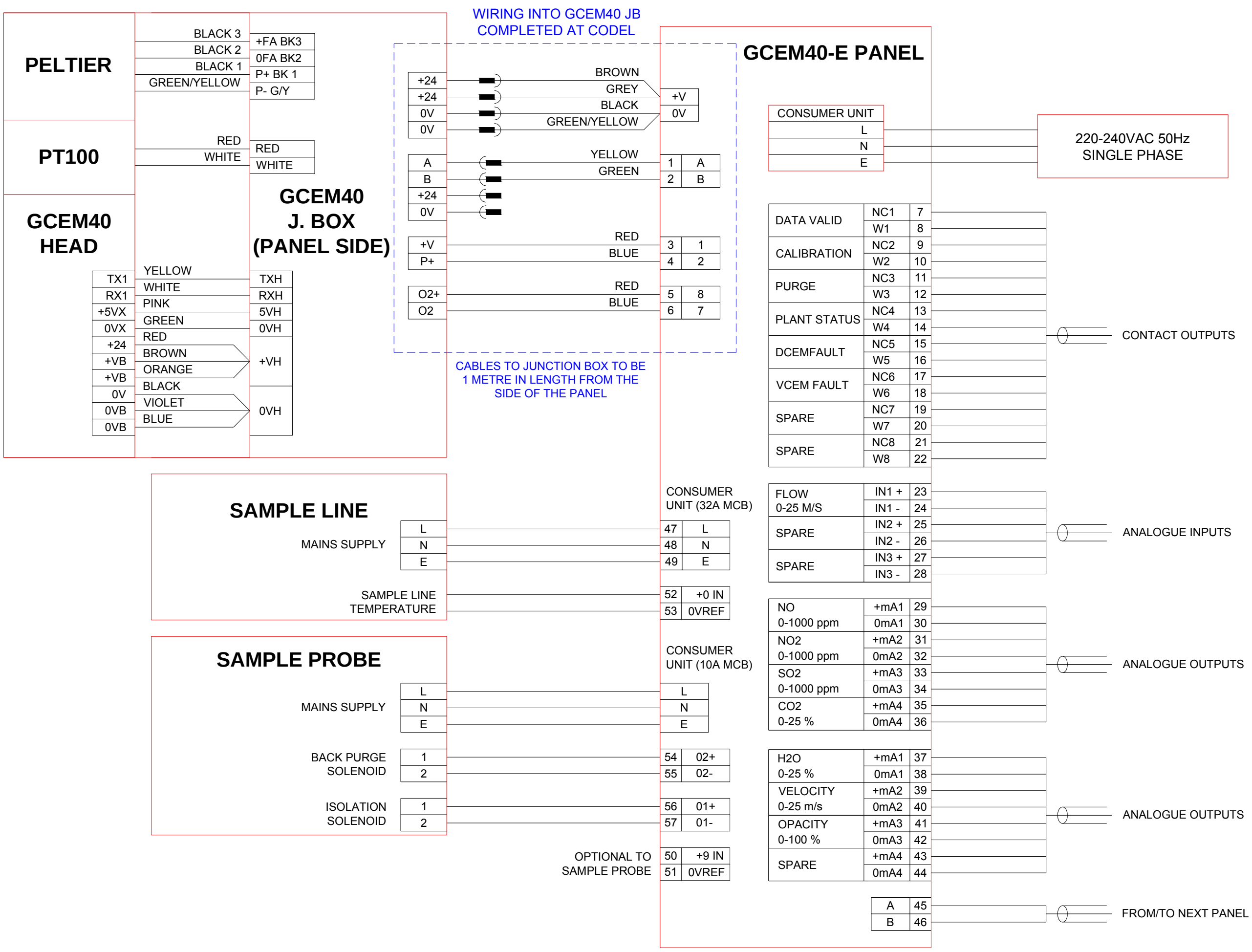


Figure 8 – System inter-connections

7.5.3 System Connection Schedule – Optional Opacity Measurement

Typical inter-connections for the system cables are shown in Figure 8.

Please refer to contract-specific drawings for accurate details.

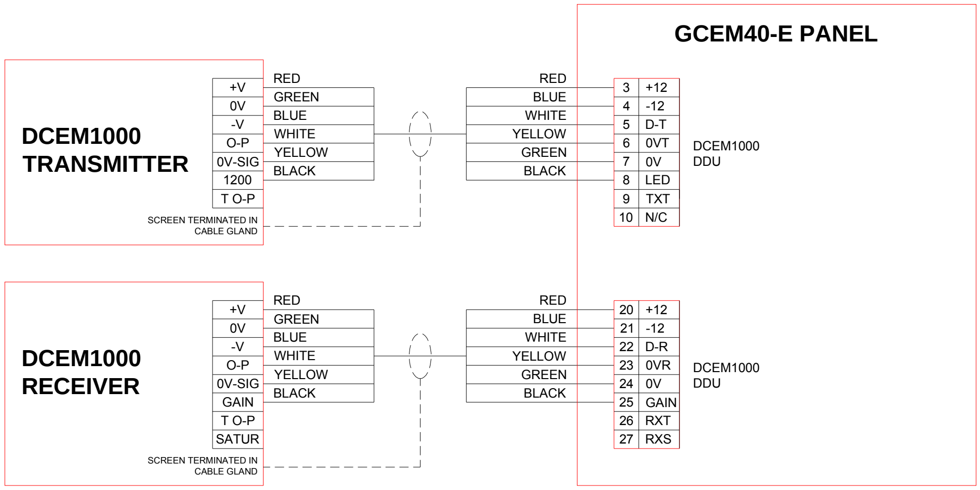


Figure 9 – Optional DCEM inter-connections

7.5.4 System P+ID

Typical pneumatic and electrical connections are shown in Figure 10 and Figure 11.

Please refer to contract-specific drawings for accurate details.

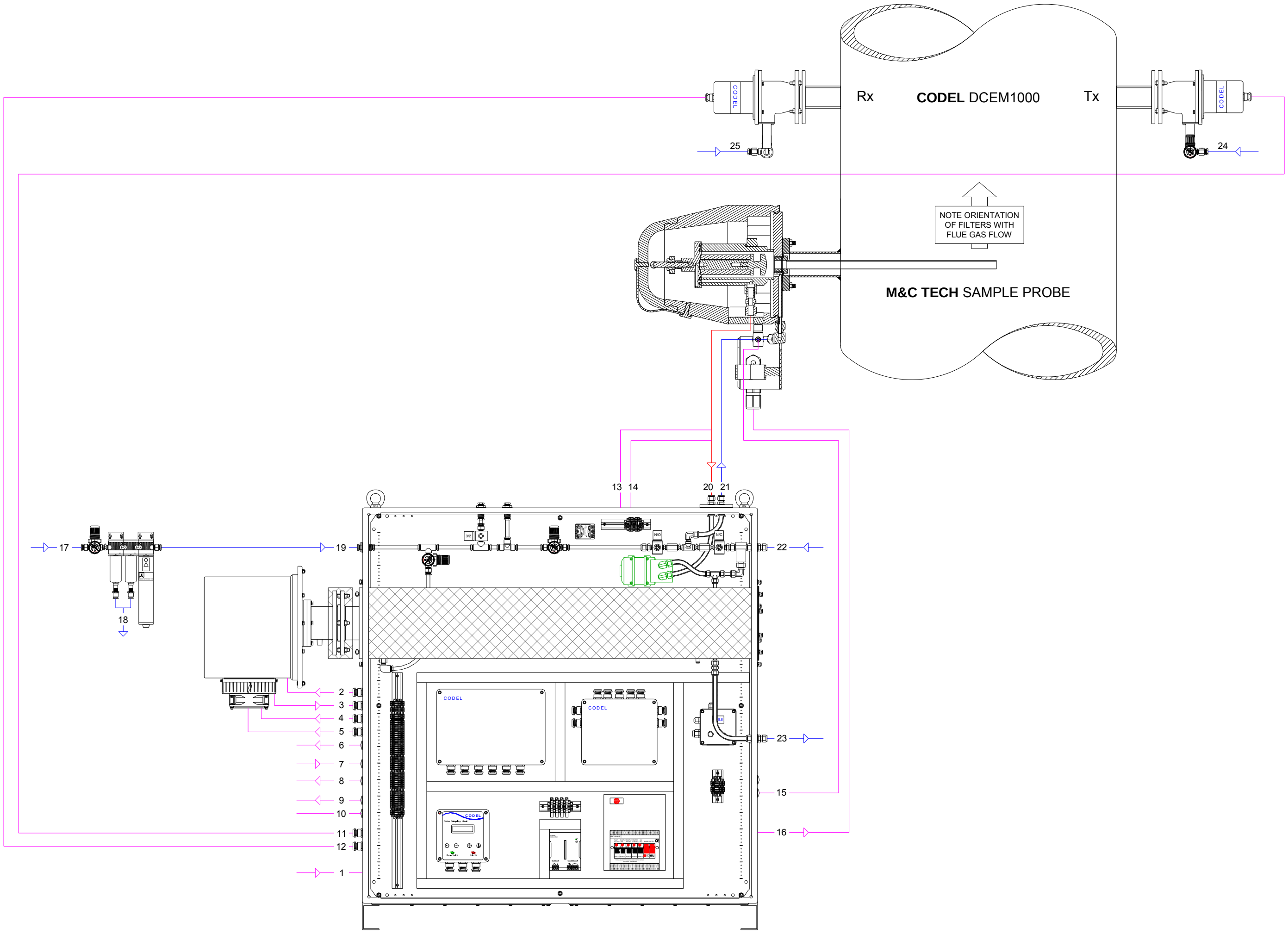


Figure 10 – P+ID schematic

#	Cable/ Pipe	Function	From		To		Minimum Specification	Supplied
1	Cable	GCEM40-E Panel Power	Clients Power	Isolator	GCEM40-E Panel	Consumer unit	3-Core Cable, 230/240V, 50Hz, 1ph, 100A	Others
2	Cable	GCEM40 Power	GCEM40-E Panel (24V PSU)	Terminals +V, 0V	GCEM40 JB	Trailing socket	1.5m 4-core 32/0.25 1.5mm2 PVC Cable	CODEL
3	Cable	GCEM40 Comms	GCEM40-E Panel (SCU)	Terms (1-2)	GCEM40 JB	Trailing Plug	1.5m DEF STAN 16-2-4C 4 Core Cable	CODEL
4	Cable	Pressure	GCEM40-E Panel	Terms (3-4)	GCEM40 JB	Terminals	1.5m 2-Core Screened Cable	CODEL
5	Cable	O2 Connection	GCEM40-E Panel	Terms (5-6)	GCEM40 JB	Terminals	1.5m 2-Core Screened Cable	CODEL
6	Cable	Contact Outputs	GCEM40-E Panel	Terms (7-22)	Clients Computer/DCS		8 x pairs 50 VDC 1A Max	Others
7	Cable	Analogue Inputs	Clients Equipment		GCEM40-E Panel	Terms (23-28)	3 x Twin Twisted Pairs 500 Ohm MAX	Others
8	Cable	Analogue Outputs 1-4	GCEM40-E Panel	Terms (29-36)	Clients Computer/DCS		4 x Twin Twisted Pairs 500 Ohm MAX	Others
9	Cable	Analogue Outputs 5-8	GCEM40-E Panel	Terms (37-44)	Clients Computer/DCS		4 x Twin Twisted Pairs 500 Ohm MAX	Others
10	Cable	RS485 To/From Next Panel	Current Panel	Terms (45-46)	Next Panel	Terms (45-46)	2-Core Screened Cable (MAX 1km)	Others
11	Cable	DCEM1000 Transmitter	GCEM40-E Panel -DDU	Terminals	DCEM1000 Transmitter	Terminals	10m 6-Core Cable	CODEL
12	Cable	DCEM1000 Receiver	GCEM40-E Panel -DDU	Terminals	DCEM1000 Receiver	Terminals	10m 6-Core Cable	CODEL
13	Cable	Sample Line Power	GCEM40-E Panel	Terms (47-49) By Others	Sample Line	Terminals	3-Core Cable, 230/240V, 50Hz, 1ph, 100A	Others
14	Cable	Sample Line Temp	GCEM40-E Panel	Terms (52-53)	Sample Line	Terminals	2-Core Cable	Others
15	Cable	Sample Probe Isolation Solenoid	GCEM40-E Panel	Terms (56-57)	Isolation Solenoid (Sample Probe)	Terminals	2-Core Screened Cable	Others
16	Cable	Sample Probe Power	GCEM40-E Panel	Consumer Unit (10A MCB)	Sample Probe	Terminals	3-Core Cable, 230/240V, 50Hz, 1ph, 100A	Others
17	Pipe	Plant Air Supply	Clients Plant Air Supply	Isolating valve	Beko Air Dryer	10mm Comp. Brass	10mm OD Nylon/PTFE	Others
18	Pipe	Autodrains	Beko Air Dryer	10mm Push Fit	Clients Drain		10mm OD Nylon/PTFE	Others
19	Pipe	GCEM40-E Dry Air Supply	Beko Air Dryer	10mm Comp. Brass	GCEM40-E Panel	10mm Comp. Brass	10mm OD Nylon/PTFE	Others
20	Pipe	Sample Gas in	Sample Probe	10mm Comp. S/S	GCEM40-E Panel	10mm Comp. S/S	10mm OD Nylon/PTFE	Others
21	Pipe	GCEM40-E Zero/Span (Cal)	GCEM40-E Panel	10mm Comp. S/S	Sample Probe (Solenoid)	10mm Comp. S/S	10mm OD Nylon/PTFE	Others
22	Pipe	Span Gas in	Gas Bottle	Isolating Valve	GCEM40-E Panel	10mm Comp S/S	10mm OD Nylon/PTFE	Others
23	Pipe	Sample Gas Outlet (Vent)	GCEM40-E Panel	10mm Comp. S/S	Clients Isolated Vent	10mm Comp S/S	10mm OD Nylon/PTFE	Others
24	Pipe	DCEM1000 Transmitter Purge	Clients Clean Plant Air Supply	Isolating Valve	DCEM1000 Transmitter Purge	10mm Comp. Brass	10mm OD Nylon/PTFE	Others
25	Pipe	DCEM1000 Receiver Purge	Clients Clean Plant Air Supply	Isolating Valve	DCEM1000 Receiver Purge	10mm Comp. Brass	10mm OD Nylon/PTFE	Others

Figure 11 – P+ID connections

7.5.5 Cable Gland Fitment

All items are supplied with cable gland entries fitted with blanking plugs. Cable glands are packed separately.

To fit a cable gland (Figure 12), remove the appropriate blanking plug, remove the locking nut from the cable gland, screw the gland into the threaded hole in the case from the outside and tighten **by hand** to compress the seal. Re-fit the locking nut from inside the case and tighten.

In the case of screened cables it is important to terminate the screen correctly at the cable gland. Unscrew the cable gland compression ring and remove the compression sleeve, rubber seal and washer and thread onto the cable to be fitted. Strip back the cable outer sheath to reveal as much of the inner cores as is needed for connection within the case, taking care not to sever the screen mesh. Strip back the screen mesh to leave sufficient screen to wrap around the cable gland washer. Insert the cable into the gland, locating the screen washer and rubber seal into the gland. Push the compression sleeve over the rubber seal and screw in the gland compression ring, tightening to provide a seal over the cable.

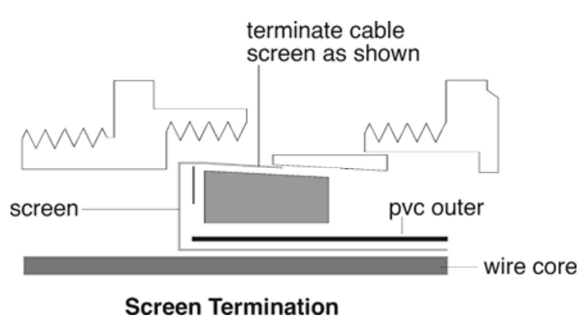


Figure 12 : Cable gland installation

8 Putting into service

8.1 Pre-Commissioning Checks

Before proceeding the following checks should be carried out.

Check all wiring and connections for conformity with the contract-specific connection schedules and that the switch on the 24VDC power supply unit is set for the correct input voltage.

Although the analyser is equipped with all practical safeguards against the consequences of incorrect wiring it is not possible to provide total protection against all errors. Damage arising from incorrect wiring will invalidate the warranty.

8.2 Compressed air ON

Check that suitable compressed air is connected and available at the inlet of the air dryer unit at a minimum pressure of 5barg (75psig).

8.3 Power ON

Open the cabinet and switch on power at the main panel isolator. The system should power up successfully with no operator intervention.

Allow 4 hours for the system to stabilise at optimum operating temperatures before continuing. Note that although measurement values will now be displayed they will probably be meaningless and will remain so until the commissioning procedure is complete.

During this initial period, the pump will NOT start up until all parts of the system are at final operating temperatures.

9 Commissioning

The gas analyser should now be fully installed. It will have been set up and tested before leaving the factory specifically for this site. However, final configuration of analogue inputs and outputs may require changes to suit site conditions.

It is advisable to calibrate the entire system after any changes have been effected.

These procedures can be carried out with the plant ON or OFF and using a separate laptop PC with CODEL SmartCom software.

The optional opacity/dust density must be aligned and zeroed with the plant off as described in the relevant manual.

Please note that this analyser cannot provide a true measurement of Dust Density until a representative Dust Factor has been established by third party isokinetic analysis.

10 Standard MODBUS Communication

Using standard MODBUS protocol function 03 allows a host to obtain the contents of one or more holding registers in the CODEL SCU.

The request frame from the host (typically a DCS or SCADA) defines the relative address of the first holding register followed by the total number of consecutive registers to be read. The response frame from the CODEL SCU lists the contents of the requested registers, returning 2 bytes per register with the most significant byte first. A maximum of 125 registers can be accessed per request.

The formats of the request and response frames are detailed below, where 'X' and 'n' are hexadecimal variables. An example of a MODBUS register map is shown below:

Host Request Frame

01	Address
03	Function code
XX XX	Address of starting register
00 XX	Number of consecutive registers
XX XX	CRC

Slave Response (from CODEL MODBUS SCU)

01	Address
03	Function code
XX	Byte count
XX XX …to… …to…	Date from starting register
XX XX	Date from nth register
XX XX	CRC

Standard baud rate - 4800 or 9600

Bits per byte - 1 start bit

11 8 data bits (least significant sent first)
12 1 stop bit
13 no parity

Data Register Locations at the CODEL GCEM-40E for Interrogation using Standard MODBUS are shown in the following table Figure 13.

REGISTER FORMAT					CUSTOMER SPECIFIC DETAILS			
Location		Analyser Data	Data	Units	Typical Range	Tag	Value	Description
0070	G-CEM40x0 channel 1	instantaneous	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0071	G-CEM40x0 channel 2	instantaneous	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0072	G-CEM40x0 channel 3	instantaneous	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0073	G-CEM40x0 channel 4	instantaneous	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0074	G-CEM40x0 channel 5	instantaneous	XXXX	1ppm (or 0.1%)	0-3000ppm (0-50.0% if CO2 or H2O)		XXXX	
0075	Not Used	instantaneous	XXXX				XXXX	
0076	Not Used	instantaneous	XXXX				XXXX	
0077	Normalisation Source - Temperature	instantaneous	XXXX	1 degree C	0-300		XXXX	
0078	Normalisation Source Data - Oxygen	instantaneous	XXXX	0.1%	0-25.0		XXXX	
0079	Normalisation Source Data - Pressure	instantaneous	XXXX	1kPa	0-160		XXXX	
007A	Normalisation Source Data - Water Vapour	instantaneous	XXXX	0.1%	0-50.0		XXXX	
007B	Flue Gas Temperature	instantaneous	XXXX	1 degree C	0-300		XXXX	
007C	Data Valid	instantaneous	XXXX	logic	XX00=valid, else=invalid		XXXX	
007D	Purging/Calibration Status	instantaneous	XXXX	logic	00XX=purge off, XX00=Cal off		XXXX	
007E	Plant Status/Head Indentifier	instantaneous	XX0A	logic	000A=plant on, else Plant Off		XX0A	
007F	G-CEM40x0 DDU Comms validation	instantaneous	XXXX	logic	00XX=valid, FFXX=no reply		XXXX	
0400	G-CEM40x0 channel 1	instantaneous	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
0401	G-CEM40x0 channel 1	instantaneous	XXXX	mg/m3 (or 0.1%)	0-9999mg/m3 (0-50.0% if CO2 or H2O)		XXXX	
0402	G-CEM40x0 channel 1	instantaneous	XXXX	mg/Nm3 (or 0.1%)	0-9999mg/Nm3 (0-50.0% if CO2 or H2O)		XXXX	
0403	G-CEM40x0 channel 2	instantaneous	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
0404	G-CEM40x0 channel 2	instantaneous	XXXX	mg/m3 (or 0.1%)	0-9999mg/m3 (0-50.0% if CO2 or H2O)		XXXX	
0405	G-CEM40x0 channel 2	instantaneous	XXXX	mg/Nm3 (or 0.1%)	0-9999mg/Nm3 (0-50.0% if CO2 or H2O)		XXXX	
0406	G-CEM40x0 channel 3	instantaneous	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
0407	G-CEM40x0 channel 3	instantaneous	XXXX	mg/m3 (or 0.1%)	0-9999mg/m3 (0-50.0% if CO2 or H2O)		XXXX	
0408	G-CEM40x0 channel 3	instantaneous	XXXX	mg/Nm3 (or 0.1%)	0-9999mg/Nm3 (0-50.0% if CO2 or H2O)		XXXX	
0409	G-CEM40x0 channel 4	instantaneous	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
040A	G-CEM40x0 channel 4	instantaneous	XXXX	mg/m3 (or 0.1%)	0-9999mg/m3 (0-50.0% if CO2 or H2O)		XXXX	
040B	G-CEM40x0 channel 4	instantaneous	XXXX	mg/Nm3 (or 0.1%)	0-9999mg/Nm3 (0-50.0% if CO2 or H2O)		XXXX	
040C	G-CEM40x0 channel 5	instantaneous	XXXX	1vpm (or 0.1%)	0-3000vpm (0-50.0% if CO2 or H2O)		XXXX	
040D	G-CEM40x0 channel 5	instantaneous	XXXX	mg/m3 (or 0.1%)	0-9999mg/m3 (0-50.0% if CO2 or H2O)		XXXX	
040E	G-CEM40x0 channel 5	instantaneous	XXXX	mg/Nm3 (or 0.1%)	0-9999mg/Nm3 (0-50.0% if CO2 or H2O)		XXXX	

Figure 13 : Modbus data locations

14 Calibration and Measurement Verification

14.1 Introduction

The following sections provide information on the analyser calibration at the time of supply (Factory Calibration), methods by which measurement/calibration can be verified and, should it be necessary, facilities for calibration adjustment.

14.2 Factory Calibration

GCEM Series analysers use very tightly controlled infrared wavebands that are specific to each of the gases measured. The scale-shape for each measurement (i.e. the relationship between concentration and output) is therefore uniquely defined and the output, at any measurement level, is only dependent on gas concentration, gas temperature and any system zero error.

Gas temperature is a continuous measured input. Any zero error detected during commissioning is corrected as part of the commissioning procedure and any subsequent zero drift is automatically detected, measured and corrected.

Factory test procedures ensure that analyser calibration is traceable to national standards and, with the inherent stability of scale-shape and automatic zero, only a significant fault (one that would be detected by the analysers' own diagnostic routines) or incorrect installation/commissioning can cause a calibration error. The analyser is equipped with means by which the factory calibration can be modified on site, but great care should be taken to ensure that calibration adjustment is absolutely necessary before proceeding with this routine as it will overwrite the factory calibration settings.

14.3 Verification

Even though GCEM Series analysers are designed to maintain their calibration stability almost indefinitely, it is often necessary to verify measurement performance. This may be a routine check after an extended period of operation, to confirm unusual measurements or as an audit requirement by some external authority.

GCEM Series analysers have a range of verification options, designed to provide varying levels of independence from both the analyser itself and CODEL standards. Which one is chosen depends on the resources available and legislative demands. The different methods can be used in combination to minimise the amount of external work required to maintain performance confidence.

14.3.1 Automatic Zero Calibration

This automatic procedure is carried out by the analyser itself and provides regular checking and, if necessary, adjustment of the zero for each measurement. The automatic zero calibration procedure provides a regular check on analyser performance without the need for operator intervention. The operation of this is covered in the IEM for Windows^{R} manual.

14.3.2 Manual Zero Verification

The zero of all measurement channels can be verified manually at any time after being powered-up for at least four hours.

The purge air must be dry to the specified level (-20°C). Any water vapour in the measurement path will show an offset not only on the water vapour measurement itself, but also on any other measurements that the analyser automatically corrects for cross-sensitivity to water vapour. Any calibration subsequently initiated to correct these apparent offsets will be incorrect.

If an analyser error is still suspected contact CODEL or the local service to establish what information to provide.

14.4 Comparative Testing

Legislation often require that continuous emissions monitoring systems are audited at regular intervals (usually once or twice per year) against an independent approved test system. These approved test systems are always extractive analysers and measurements must normally be carried out by organisations independent of the CEMS manufacturer or the plant operator.

If the analyser is working correctly and the comparative testing is carried out to the prescribed procedure, it is rare for such exercises to be other than a routine confirmation of performance. However, to ensure that any discrepancies can be easily resolved, it is recommended that the following precautions be taken:

a) Check what level of agreement is required by the relevant legislation. Most legislation is realistic about what can be expected in comparing a system that provides an average of a significant percentage of the whole duct or stack and one that averages a number of discrete measurements from different positions at different times.

b) Be aware of the specified procedure for the comparative test and ensure that appropriately qualified personnel using certified equipment and test gases carry it out. **Do not accept the validity of quick, single-point checks with uncertified analysers.**

c) Ensure that, when comparing measurements made by the GCEM and the audit system, the measurements are on the same basis, i.e.

- they have been recorded simultaneously and during a period when plant operation is stable
- the measurement averages are equivalent (e.g. 10-minute rolling average measurements from the Model 4000 should be compared with 10-minute averages from the audit system and not measurements averaged over significantly different time periods).
- all measurements are normalised/corrected to the same standard conditions of temperature, pressure, water vapour (dry or wet) and excess oxygen (or CO_2).

d) If the reasons for any discrepancies are not apparent it is strongly recommended that CODEL or its local service agent be consulted before any attempt is made to recalibrate the Model GCEM analyser. Re-calibrating to a reference measurement that is subsequently found to be unreliable creates many unnecessary difficulties. **DO NOT RECALIBRATE UNLESS ABSOLUTELY CERTAIN THAT IT IS NECESSARY!**

14.5 Re-calibration

The laws of physics determine that the scale shape (i.e. the relationship between gas concentration and analyser output) is absolutely fixed for a defined measurement waveband. Only serious faults, (e.g. a leaking gas cell) that would be detected and registered as a data invalid condition by the analyser itself, can modify this relationship.

If re-calibration cannot wait until the next scheduled automatic zero calibration (Section Automatic Zero Calibration) a manual zero calibration may be implemented (Section Manual Zero Verification).

Any discrepancy that is now outside the combined measurement uncertainties of the test gas (or comparative measurement) and the analyser, can only be due to either a significant fault in the analyser itself, inaccuracy in the test gas or inappropriate comparative testing.

Consider and check all other possibilities before assuming an analyser fault. Analyser faults are rare, are usually detected by the analyser itself (initiating an automatic data invalid state) and likely to produce very large errors. When convinced that any discrepancy is due to an analyser problem contact CODEL or the local service agent to determine what information to provide.

15 Maintenance – GCEM

This version of the GCEM series requires no routine maintenance other than care of the air purge supplies.

Ensure that all filters, dryers and automatic drain units are serviced and maintained according to the manufacturers' instructions.

Ensure that the compressor is properly serviced and does not inject more oil into the air supply than the filter system can remove.

THE ACCURACY AND TROUBLE-FREE OPERATION OF THE ANALYSER DEPENDS ON THE PURGE AIR SUPPLIES BEING MAINTAINED TO THE REQUIRED STANDARDS.

The probe and fittings are designed, by choice of materials and construction, to be corrosion resistant and to provide many years of trouble-free operation. However, it is good practice to take the opportunity of plant shutdown periods to visually examine components that are not accessible during normal operation.

The heated pump and sample probe will need to be serviced in line with the manufactures recommendations.

The equipment is designed to keep the levels of maintenance to an absolute minimum.

16 Maintenance – Optional DCEM

16.1 Cleaning Windows

It is important that the optical windows of both the transmitter and the receiver be kept reasonably clean and any mounting tubes free from build-up of dust and fly ash. This can be accomplished by removing the transmitter and receiver and wiping their windows with a soft, dry cloth. If the windows are cleaned, a calibration is advised afterwards.

16.2 Clean Flue Condition Available

Should a clean flue condition become available and the instrument was commissioned with an opacity offset, it is recommended that a calibration be conducted to take advantage of this situation.

17 Fault Finding

CODEL GCEM and DCEM analysers are sophisticated devices and any problems necessitating internal repair or adjustment should only be undertaken by fully trained technicians.

In the absence of any CODEL trained technicians on site, it is strongly recommended that, in the event of a fault, CODEL or its local service agent be contacted immediately with equivalent current information.

18 List of Figures